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Deep Learning in Smart Healthcare: A GAN-based Approach for Imbalanced Alzheimer's Disease Classification

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Abstract

Alzheimer's disease (AD) is a type of dementia that leads to memory loss and impairment, which affects patients' lives badly. It is not curable yet, but its progression can be slowed down if detected at earlier stages. In this research study, we propose a transfer learning-based convolutional neural network (CNN) model to classify magnetic resonance imaging (MRI) into one of four stages of Alzheimer's disease. One of the major limitations of the deep learning-based classification model is the non-availability of healthcare datasets related to AD. The widely used Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset has a major class imbalance issue. We propose a generative adversarial network (GAN) based data augmentation technique to overcome the data imbalance. This promotes the investigation of applying GANs to generate synthetic samples for minority classes in Alzheimer's disease datasets to enhance classification performance. The results show the progression in the overall classification process of AD.

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Keywords: Alzheimer disease (AD), Deep learning, Convolutional Neural Network (CNN), Computer-aided diagnosis (CAD)

1. Introduction

Healthcare is undergoing a transformative shift from traditional methodologies to smart, technology-driven approaches, largely fueled by advancements in artificial intelligence (AI), machine learning, and deep learning [1]. These

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technologies are revolutionizing the field by enabling more precise diagnostics, personalized treatment plans, and improved patient outcomes. AI algorithms rapidly analyze vast amounts of data, uncovering patterns that may not be visible to human clinicians [2]. Machine learning models adapt and improve over time, learning from new data inputs to make more accurate predictions and recommendations. Deep learning further enhances capabilities, particularly in image recognition tasks crucial for diagnosing diseases from medical imaging. This shift towards smart healthcare is optimizing clinical workflows and making healthcare more proactive and preventive, significantly enhancing the efficiency and quality of care provided to patients [3].

Alzheimer's disease (AD) is the leading type of dementia, characterized by a gradual decline in brain function, primarily observed in older individuals [4]. Compared to the patient's prior state of functioning, there is an indication of a decline in memory and other cognitive skills, which is symptomatic of primary dementia syndrome. Alzheimer's disease causes about two-thirds of cases of progressive cognitive decline in the elderly population, with an estimated prevalence of over 130,000 individuals [4]. According to data from 2006, the global prevalence of Alzheimer's disease (AD) was estimated at 26.6 million individuals. It is projected that this number will increase twofold every two decades. Researchers predict that by the year 2046, Alzheimer's disease will affect about 1.2% of the global population [5]. The identification of mild cognitive impairment (MCI) is primarily linked to the early detection of Alzheimer's disease, serving as an early stage of the condition.

While the cognitive impairments and memory complaints associated with MCI do not significantly impact the individual's everyday functioning, there is research supporting that there is a significant chance of MCI progressing to AD or other types of dementia [6]. The timely and accurate identification of Alzheimer's disease, particularly the progression of MCI to AD, provides AD patients with an understanding of the severity of their condition and enables them to implement preventive measures such as changing their lifestyle and taking medication [7].

Diagnosing Alzheimer's disease within its initial stages creates challenges, yet accurate predictions that could reduce the extent of its effects. Neuronal degeneration and subsequent cell death in the brain lead to the development of total amnesia. According to reports, this illness affected about 50 million people worldwide in 2017. The progression of Alzheimer's disease can be categorized into four distinct stages, which are outlined as follows:

- **Pre-dementia or mild cognitive impairment** refers to a stage of cognitive decline. This phase occurs before the onset of the illness. Short-term memory loss is a cognitive impairment sometimes misinterpreted as a typical manifestation of the aging process;
- **Early Stage Dementia** In the early stages, the capacity to develop and store new memories may be reduced, although the ability to keep older memories is preserved;
- **Advance Stage Dementia** This phase is beyond its initial phase and will begin to have several effects on an individual's life. The impairment of motor function will hinder the ability to move. Individuals may experience speaking, memory retention, and reading and writing challenges;
- **Last Stage Dementia** This represents the ultimate phase of the disease, resulting in eventual death.

While Alzheimer's disease may not directly result in death rates, the presence of diseases such as infection or pneumonia could worsen the individual's overall physical well-being. During this phase, the individual will depend primarily on the caretaker.

The primary aim of our study is to identify and diagnose Alzheimer's disease in its initial stages. We propose a methodology for determining whether a person has Alzheimer's and further classify the disease into the four stages mentioned previously. We take magnetic resonance imaging (MRI) from a scanner and then preprocess the image for normalization and smoothing to remove anomalies. Afterwards, we divide the MRI image into black and white pixels. Our CNN will extract the features from the image to classify dementia into four categories: early stage, moderate stage, advanced stage, and Last Stage. The Dataset of Alzheimer's Disease Neuroimaging Initiative (ADNI) [7] has been used for this proposed research. Further, this paper is also applying GAN [8] to solve the issue of a highly unbalanced dataset of Alzheimer's.

2. Literature Review

Alzheimer's is a neurodegenerative condition characterized by the gradual death of neuronal cells, leading to memory loss, cognitive functions, behavioral changes, and language skills deterioration. Approximately 5.4 million individuals in the United States are affected by this condition, with 5.2 million of them being elderly people [9]. Numerous research studies [10, 11] have emphasized the importance of quickly and accurately diagnosing Alzheimer's disease. Early diagnosis allows for timely treatment, appropriate symptom management, implementation of safety precautions, and cost-effective therapies. Research in Alzheimer's disease diagnosis has explored various computer-aided multi-class diagnosis methodologies, achieving varying levels of success. However, most studies focusing on computer-aided diagnosis for Alzheimer's disease predominantly address binary classification, that is, determining whether a patient is diagnosed with Alzheimer's or not. In this context, [12] provides an extensive review of the current methodologies used in computer-aided diagnosis systems (CADS) for Alzheimer's disease (AD), employing neuroimaging techniques. While this work offers significant insights, further research is necessary to develop more accurate and reliable CADS methodologies for AD diagnosis.

Dementia is a broad term that includes a variety of cognitive impairments affecting memory, thinking, and behavior. Alzheimer's disease (AD) is the most prevalent form of dementia, representing about 60-80 percent of cases [13]. AD is marked by a progressive and irreversible decline in cognitive function, significantly affecting independence and quality of life [14].

The methodologies for dementia diagnosis vary and include axial tomography (CT) [15], structural imaging using magnetic resonance imaging (MRI) [16], and positron emission tomography (PET) [17]. Single-photon emission computed tomography (SPECT) is another technique, using gamma rays emission from a radiopharmaceutical to create three-dimensional images of the brain's internal structures and functions [18]. MRI is currently utilized to examine brain morphology [19], with several sequences available, such as T1-weighted imaging, T2-weighted imaging, and fluid-attenuated inversion recovery (FLAIR) imaging [19]. The most commonly used sequences in medical imaging are T1-weighted (T1-w) and T2-weighted (T2-w) scans, with FLAIR being the third frequently used sequence [19]. Diffusion-weighted imaging (DWI) and diffusion tensor imaging (DTI) are applied to detect diffusion-related phenomena [20]. Studies often involve the use of various image modalities, including structural magnetic resonance imaging (sMRI), fluorodeoxyglucose-positron emission tomography (FDG-PET), single-photon emission computed tomography (SPECT), diffusion tensor imaging (DTI), and arterial spin labeling combined with DTI datasets (ASL + DTI) [9]. Different image modalities allow observers to detect certain phenomena that may not be visible in other modalities [9]. According to their study, the authors mainly used structural magnetic resonance imaging (SMRI), focusing on T1-weighted (T1-W) images [15]. SMRI has proven effective in diagnostic applications by clearly delineating the contrasting characteristics of healthy and pathological soft tissues within organs [16]. The dataset in this work was sourced from the PET (positron emission tomography) database [21]. The study identified support vector machines (SVM) as the most effective classifier, achieving an accuracy rate of 90.91% [21]. However, a limitation of this study is its focus solely on the binary categorization of Alzheimer's disease (AD), neglecting the potential for multi-classification of Alzheimer's disease [21].

3. Proposed Methodology

We introduced and developed a novel hybrid deep-learning model for detecting Alzheimer's disease, combining InceptionV3 and a CNN model. The overall workflow is illustrated in Figure 1, outlining the following key steps:

A: Our approach begins with gathering data from the ADNI database, consisting of 6,400 images spanning four distinct stages of Alzheimer's disease. Each stage represents a different progression of the disease. Real-world data often comes with noise and variations in image size. We apply preprocessing steps to make it suitable for deep learning, including resizing the image and feature scaling;

B: The ADNI dataset has an uneven distribution of images for different stages of Alzheimer's disease. To overcome this imbalance, we employ Generative Adversarial Networks (GANs) to generate a supplementary dataset for various Alzheimer's disease classes;

C: In this phase, we construct a hybrid model by integrating the InceptionV3 model with a tuned CNN model for Alzheimer's detection. The image dataset is then split into training and test sets. The model undergoes training and validation using appropriate images.

D: The effectiveness of our hybrid model is assessed by evaluating its accuracy. Performance metrics such as accuracy, precision, recall, f1 score, and confusion matrix are employed in this stage of the proposed model. Additionally, we conduct a comparative analysis with existing models to demonstrate the efficacy of our approach.

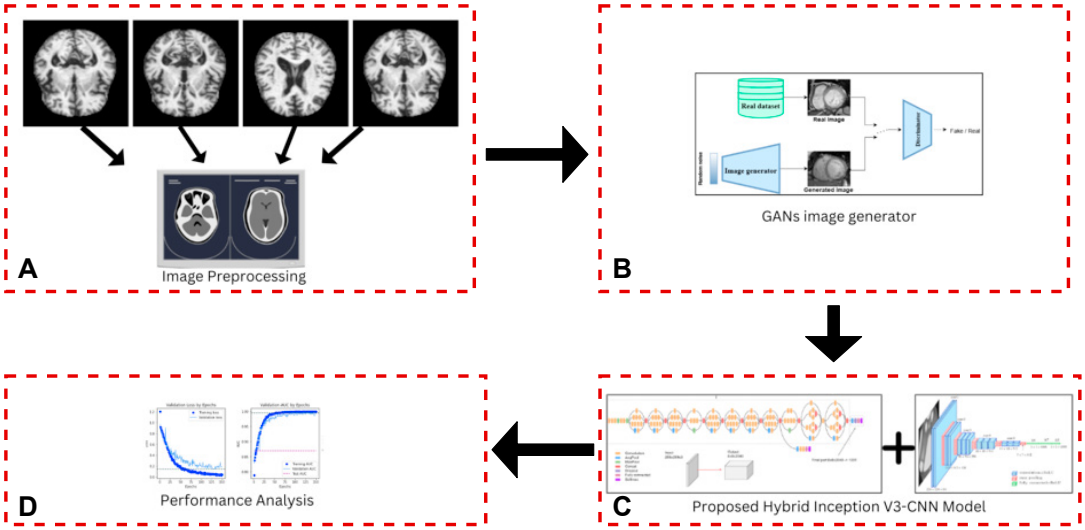


Fig. 1. Block Diagram of the proposed methodology.

3.1. Dataset Details

The Alzheimer's Disease Neuroimaging Initiative (ADNI) [7] is a longitudinal multi-site observational research of older people with normal cognition, mild cognitive impairment (MCI), or AD. The National Institutes of Health (NIH) and industry support it jointly through the NIH Foundation. The purpose of this research is to compare the effectiveness of using information obtained from magnetic resonance imaging (MRI), fludeoxyglucose positron emission tomography (18F) (FDG PET), urine, serum, and cerebrospinal fluid (CSF) biomarkers, as well as clinical and neuro psychometric assessments, to track disease progression in the three groups mentioned above of the elderly people. The 55 collaborating centers in North America collect imaging, clinical, and biologic samples from 200 cognitively normal seniors, 400 individuals with MCI, and 200 individuals with AD across multiple time points. Table 1 summarizes the details of the ADNI database constitution.

Table 1. ADNI Dataset details for multi-class classification

Class Label	No. of Samples
Non demented	3200
Very mild demented	2240
Mild demented	896
Moderate demented	64

3.2. Data Pre-processing

Data pre-processing involves a set of computerized methods to prepare input data for the optimal functioning of learning models. It plays a crucial role in refining the data by eliminating noise, handling missing values, and

structuring it to create a final image that is as clean as possible, containing only relevant information for the intended task. The pre-processed dataset significantly contributes to the effectiveness of training and testing the entire proposed research model. This hybrid model employs three types of data pre-processing: Image Filtering, data normalization, and data augmentation. Figure 1, in the rectangle A, illustrates the sequence of these pre-processing techniques.

3.3. Data Augmentation using Generative Adversarial Networks (GANs)

The presented research introduces a novel Generative Adversarial Network (GAN) architecture [22], comprising a discriminator, a generator, and their combined network. Each model is meticulously designed to accomplish specific tasks to generate realistic MRI images of Alzheimer's disease. The models' architectures are outlined below, providing a detailed insight into their layers and parameters.

The discriminator employs a sequence of convolutional and dense layers interleaved with leaky rectified linear unit (LeakyReLU) activations. The network culminates in a dense layer producing a single output, representing the discriminator's binary decision on the authenticity of an input image. The generator model, responsible for synthesizing images, starts with a dense layer, followed by LeakyReLU activation and reshaping to transform the initial noise vector. Subsequent transposed convolutional layers incrementally upsample the data until a final convolutional layer produces a single-channel image. The combined network encapsulates the generator and discriminator, as this model integrates them sequentially with an input layer for the initial noise vector. The generator aims to produce compelling synthetic images, while the discriminator distinguishes between genuine and synthetic images. The combination of these models forms a GAN that undergoes training to refine both the generation and discrimination processes, ultimately yielding enhanced image synthesis capabilities. Figure 2 presents some examples of the original images used and some of the GAN generated. This part of the algorithm is depicted in the Figure 1, in the rectangle B.

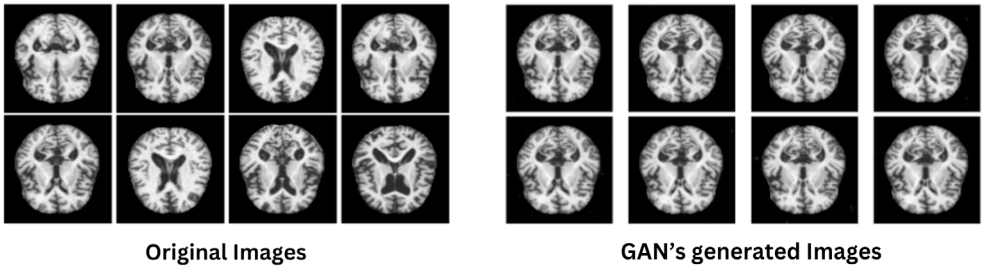


Fig. 2. Original vs GAN generated images.

3.4. Proposed Deep Learning Model

In our proposed hybrid model for Alzheimer's disease detection, we've utilized Inception V3, a pre-trained neural network on the ImageNet dataset. This network is excellent at extracting valuable information from MRI images. Inception V3 is renowned for its success in various computer vision tasks like image classification, object detection, and recognition. It stands out because of its ability to capture features at different scales using inception modules, which are like layers with different filters. We chose Inception V3 for its exceptional accuracy across diverse datasets and tasks, making it ideal for early Alzheimer's detection. Despite its complexity, Inception V3 is computationally efficient and optimized for real-time applications.

The Keras module applications used the Inception V3 with pre-trained weights from ImageNet. We then added 16 CNN layers, including a final output layer with four units for sorting (AD/NAD/MAD/VMAD), as depicted in Figure 1, in the rectangle C. During the initial training phase, we froze all layers of the base model and only trained the manually added CNN layers for 100 epochs. The CNN layers are crucial building blocks of our model. Convolution layers apply filters to inputs, creating activation and feature maps that reveal patterns in the input. Pooling layers reduce feature set proportions, making the network more efficient. Our CNN model consists of 16 layers, including dropout layers to prevent overfitting, a pooling layer, batch normalization layers, dense layers with ReLU activation, and a final dense layer for predictions using softmax.

The efficiency of our proposed model surpasses existing models in classifying Alzheimer's disease from MRI images. However, training the hybrid model in our Microsoft Windows environment takes approximately 10 to 12 hours. This part of the algorithm is presented in Figure 1, in the rectangle D.

4. Experimentation and Results

Different evaluation metrics can be used to prove the proposed system's performance. The classification algorithm divides the data into four categories: True positive (TP) for the correctly classified true labels, True negative (TN) for correctly classified false labels, False positive (FP) for the incorrectly classified false labels, and False negative (FN) incorrectly classified true labels. Precision, recall, F1-score, and support are metrics used in machine learning to evaluate the performance of classification models. Precision measures the accuracy of positive predictions, indicating how many predicted items are positive. Recall measures the classifier's ability to find all positive samples, indicating how many positives the model can capture. The F1-score provides a single metric to assess the balance between them, especially in imbalanced datasets. Support, the number of actual occurrences of a class in the dataset, helps identify imbalances that might influence model performance. These metrics provide a comprehensive overview of classification model performance.

4.1. Experimental Setup

We conducted various experiments to train and evaluate our model (depicted in Figure 1 C part. Initially, we trained the model without augmenting the dataset, using the RMSprop optimizer with a learning rate of 0.0001 and the Adam optimizer with the same learning rate. Subsequently, we enhanced the dataset using a Generative Adversarial Network (GAN). The model underwent another round of training and testing, this time using Adam optimizers.

The generator model is responsible for creating images by leveraging the randomness of random noise, also known as the Latent Vector. This Latent Vector is obtained by sampling from a Normal Distribution, serving as the input for the Generator. Through the application of Transposed Convolution, the latent vector undergoes a transformation that generates an image. A key aspect of our model involves utilizing three Conv2DTranspose layers, which play a pivotal role in producing images by focusing on features. The discriminator is the model distinguishing between generated images, classifying them as authentic or fake. Our ultimate objective for this research is to develop a highly effective Generator that challenges the Discriminator's ability to differentiate between real and fake images.

We employ a straightforward Convolutional Neural Network featuring two Conv2D layers linked to a Dense output layer to achieve this. Given that the discriminator functions as a binary classifier, the activation function for the output layer is set to Sigmoid.

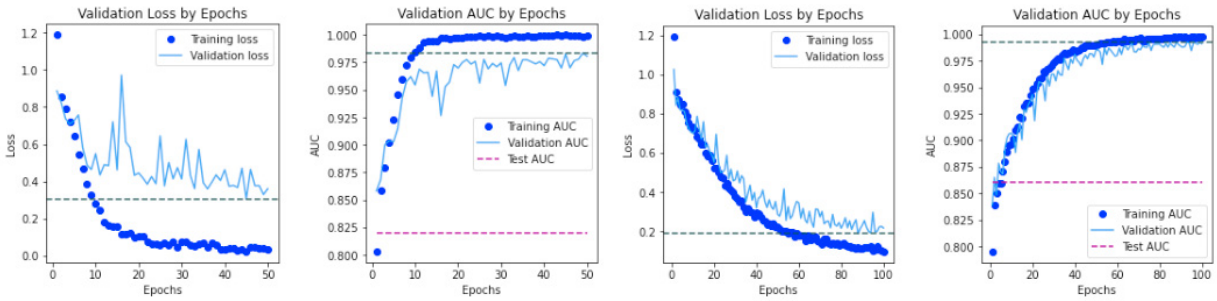
4.2. Result Analysis

In figure 3 a), the left graph illustrates that while training loss consistently decreases, reflecting the model's ability to learn from the training data, the validation loss demonstrates initial fluctuations before stabilizing. This pattern suggests some challenges in model generalization during early epochs, which are overcome as training progresses. The right graph shows the AUC values for training, validation, and test sets. Here, the training and validation AUCs converge to a high level, indicative of excellent model prediction capabilities. However, the test AUC lags slightly behind, indicating potential overfitting to the training data despite high overall accuracy. This insight is crucial for further tuning to enhance model robustness across unseen data sets. The model demonstrated losses of 0.05 for training and 0.30 for validation and high AUC values of 0.82 for training and 0.98 for validation in classifying the images in one of four stages of Alzheimer's disease.

In figure 3 b), the left graph depicts a consistent decrease in both training and validation loss, indicating that the model progressively learns and minimizes error over time, with the validation loss closely following the training loss without significant divergence, showcasing the model's stable generalization capabilities. The AUC values for training, validation, and test datasets are illustrated on the right. Both training and validation AUC improve rapidly and plateau around 0.975, signifying that the model achieves high discriminative performance. Notably, the test AUC remains slightly lower, which might suggest some overfitting despite the overall robustness of the model across different data

splits. This consistency between training and validation performance reaffirms the model's reliability and effectiveness in handling unseen data. The model demonstrated low losses of 0.05 for training and 0.19 for validation and high AUC values (close to 1) for training and 0.99 for validation, indicating its effectiveness in classifying the images in one of four stages of Alzheimer's disease.

Upon testing, the model yielded an overall AUC of 87.0%. However, a more detailed examination, as shown in Table 2, using the classification report highlighted variations in the precision, recall, and F1 scores across different classes. While the model performed well in classifying 'Normal' cases (precision: 0.71, recall: 0.83), it didn't perform well with 'Mild' and 'Very Mild' instances. The overall accuracy was 68%, suggesting a moderate performance. To enhance the model, fine-tuning parameters, addressing class imbalances, and exploring additional metrics are recommended.



(a) (b)

Fig. 3. Model evaluation: (a) Before data augmentation; (b) After data augmentation.

Table 2. Result metrics before and after data augmentation.

	Before data augmentation				**After data augmentation**			
	precision	**recall**	**F1-score**	**support**	**precision**	**recall**	**F1-score**	**support**
normal	0.62	0.83	0.71	640	0.71	0.83	0.77	640
very-mild	0.55	0.48	0.51	448	0.66	0.58	0.62	448
mild	0.62	0.12	0.20	179	0.60	0.45	0.51	179
moderate	0.00	0.00	0.00	12	0.36	0.33	0.35	12

5. Conclusion

The study effectively illustrates the potential of generative adversarial networks (GANs) in conjunction with transfer learning-based convolutional neural network (CNN) models for classifying Alzheimer's disease (AD) stages from magnetic resonance imaging (MRI) data. Despite the substantial class imbalance in the Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset and the scarcity of healthcare datasets, the suggested GAN-based data augmentation technique successfully creates synthetic samples for minority classes, improving classification performance.

By understanding that early diagnosis of AD can slow down the progression of the disease by taking care of the patient and leveraging the strengths of deep learning models, we propose Inception V3 and CNN-based solutions for the multiclass classification of AD. Using the advantages of both the CNN and Inception V3 models, the overall classification accuracy for the various stages of AD improved, which is one of the key findings. With an overall AUC of 87%, the novel approach to addressing the class imbalance issue shows promising early results, implying that more training and optimizations may produce even better results.

This research presents a viable path for the early detection and categorization of Alzheimer's disease by utilizing cutting-edge machine learning methods to address major obstacles in medical image interpretation. Future research on AD should explore advanced machine-learning techniques and integrative data analysis. Experimenting with deep

learning architectures like ResNet, DenseNet, and EfficientNet can improve feature extraction and classification accuracy. Integrating multi-modal data sources like PET scans, biomarkers, and genetic information can provide a holistic view of the disease. Longitudinal data monitoring and federated learning can enhance personalized medicine and collaborative research.

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